

Formal Models of Computational Interaction

Toward a Mathematical Foundation

Cross-disciplinary foundational research paper

Abstract

This paper evaluates whether one established formalism can represent the interaction theory developed across prior work on ontology, semantics, representation, projection, opportunity, behavior, measurement, similarity, transfer, and strategy. Labeled transition systems, Petri nets, process algebras, statecharts, event structures, workflow nets, Markov decision processes, partially observable Markov decision processes, category-theoretic process models, and multi-agent formalisms are compared by a common set of obligations: bounded interaction episodes, reciprocal coupling, partial-order behavior, typed opportunity, projection and observation, latent strategy, uncertainty, empirical traces, composition, and typed equivalence.

No candidate satisfies every obligation without extension. Labeled transition systems provide the smallest operational core and the broadest verification ecosystem, but interleaving semantics obscures true concurrency and standard forms omit observation, policy, measurement, and empirical uncertainty. Event structures preserve causality, conflict, and concurrency but are less convenient for state-rich, stochastic, and instrumented systems. Petri nets and workflow nets excel at concurrency, resources, and completion. Process algebras provide the strongest native composition and behavioral equivalence theory. MDPs and POMDPs provide decision, uncertainty, belief, and strategy semantics but ordinarily presuppose a fixed action/state model and weakly represent organizational and representational structure. Statecharts provide hierarchy and practical specification but lack one universally accepted mathematical semantics. Category theory supplies a principled language of composition and translation rather than a sufficient empirical model by itself.

The evidence therefore supports a conservative hybrid foundation rather than a new calculus: an observation-enriched probabilistic labeled transition system as the canonical semantic kernel; event-structure or Petri-net views for concurrent causality and resource flow; process-algebraic operators for composition and equivalence; and POMDP-style belief and policy layers where agency under uncertainty is part of the explanandum. A trace-and-measurement morphism connects formal executions to empirical records. The paper supplies a formal ontology mapping, core semantics, model comparison matrix, expressiveness analysis, and recommended foundation.

1. Problem Statement

The established interaction theory imposes obligations that exceed ordinary interface-state modeling. An interaction episode has declared participants, boundary, timescale, coupling, preconditions, transitions, invariants, evidence, failure, recovery, and completion. Behavior is a bounded attributed trajectory under an observation model rather than raw telemetry. Opportunity distinguishes domain-valid, capable, permitted, enabled, reachable, exposed, signaled, believed, executable, successful, and reversible action. Strategy is a latent, temporally extended organization of conditional action selection. Measurement connects records to variables and claims through explicit operationalization and uncertainty. Similarity is typed rather than universal.

A formal foundation must preserve these distinctions while remaining analyzable and compositional. The central problem is not which formalism is most expressive in the abstract. It is which minimal combination represents the established ontology without encoding every distinction as an informal annotation or inventing a new mathematical language.

2. Evaluation Method

Each candidate formalism was evaluated against the same representation obligations. The assessment distinguishes native expressiveness from expressibility by arbitrary encoding. A universal programming language can encode nearly any finite model, but such encodability does not make its semantics transparent or its analyses appropriate.

The criteria were: (1) episode boundaries and identity; (2) state and transition semantics; (3) reciprocity and coupling; (4) concurrency and causal independence; (5) nondeterminism; (6) probability and uncertainty; (7) partial observability and projection; (8) opportunity and authorization; (9) policy and strategy; (10) empirical trace alignment; (11) composition and decomposition; (12) equivalence and refinement; (13) verification and quantitative analysis; and (14) suitability for estimation from observed interaction data.

The review prioritized foundational sources: labeled transition systems and model checking; Murata on Petri nets; Milner on process algebra; Harel on statecharts; Winskel on event structures; workflow-net soundness; Sutton and Barto on MDPs; Kaelbling, Littman, and Cassandra on POMDPs; and categorical process semantics. The conclusion is comparative and architectural, not a claim that all systems should be implemented using one notation.

3. Formal Obligations Derived from the Theory

The formal object must distinguish six kinds of state: domain state, participant state, authority state, projection state, observer or belief state, and measurement state. Combining them into one Cartesian state is mathematically possible but destroys explanatory modularity and causes state-space explosion. The foundation should therefore permit factored state and typed transition labels.

An interaction episode requires more than a transition path. It requires a boundary predicate, participant assignment, coupling relation, initiating and completion conditions, and evidence linking the modeled episode to observations. Reciprocal constraint need not mean simultaneous bidirectional transition. It can be represented by causal dependence across a bounded execution: an event of one participant changes the enabled, probable, observable, or valued future of another, whose subsequent event recurrently constrains the first or the shared system.

Opportunity requires multiple predicates over actions and states rather than one action set. Projection and observation require mappings from domain states and transitions to exposed operations and observable symbols. Belief requires probability or possibility over hidden states. Strategy requires a policy-like object over histories, observations, or beliefs, plus temporally extended organization where needed. Empirical use requires a stochastic or uncertain observation channel from executions to records.

4. Labeled Transition Systems

A labeled transition system (LTS) consists minimally of states, action labels, and a transition relation. It provides the clearest common denominator for enabled actions, executions, traces, reachability, safety, liveness, refinement, simulation, and bisimulation. Interaction episodes can be represented as bounded

executions satisfying initiation and completion predicates. Typed labels can identify actor, channel, operation, authorization result, and outcome.

The principal strength of LTS semantics is analyzability. Model checking, equivalence checking, reachability, temporal logic, and compositional reductions are mature. LTSs also serve as semantic targets for statecharts, process algebras, workflow systems, and many programming languages.

The limitations are equally important. Standard interleaving semantics represents concurrent events as alternative sequences and therefore loses causal independence unless enriched. Ordinary LTSs do not natively distinguish projection from domain state, observation from execution, permission from enablement, policy from transition availability, or empirical records from formal traces. Probability, time, cost, and partial observability require extensions. The LTS is therefore the best semantic kernel, not a complete interaction model.

5. Petri Nets and Workflow Nets

Petri nets represent distributed state through places and tokens and behavior through transition firing. They are particularly strong for concurrency, synchronization, resource consumption, conflict, invariants, boundedness, liveness, and reachability. Their graphical and algebraic structure makes them well suited to workflows, multi-party protocols, and interactions in which independent activities and shared resources matter.

Workflow nets add distinguished source and sink conditions and support soundness questions concerning completion, residual tokens, and dead transitions. These concepts correspond naturally to interaction initiation, completion, deadlock, and cleanup.

Standard place-transition nets remain weak for rich data, hierarchical projection, belief, policy, and measurement. Colored, timed, stochastic, and inhibitor-arc variants increase expressiveness but fragment analysis and can approach general programming-language complexity. Petri nets should therefore be treated as a concurrent-resource view of the canonical model, especially where partial-order structure and completion are central.

6. Process Algebras

Process algebras such as CCS, CSP, and the pi-calculus represent systems through processes and operators for action prefix, choice, parallel composition, synchronization, hiding, relabeling, and recursion. Their major contribution is not simply another transition notation; it is a compositional theory in which the behavior of a whole can be derived from interacting parts and compared through rigorously defined observational relations.

This is highly aligned with the interaction theory. Participant processes can be composed through channels, internal actions can be hidden, projections can be modeled by abstraction or renaming, and typed equivalences can express what an observer can distinguish. Process algebra offers the strongest native account of compositional interaction identity and substitutability.

Classic process algebras are less natural for quantitative human behavior, empirical traces, continuous variables, resources, belief, and learning. Probabilistic, timed, stochastic, and hybrid variants exist, but no single process calculus has become the universal cross-disciplinary foundation. Process algebra is best used to define composition, interface contracts, and equivalence over the transition kernel.

7. Statecharts

Statecharts extend state machines with hierarchy, orthogonality, and broadcast communication. These features address practical complexity: hierarchical states support abstraction; orthogonal regions represent concurrency; history supports resumption; and event-triggered transitions support reactive behavior. They are highly usable for engineered interactive systems and can align naturally with interface modes and lifecycle states.

Their weakness as a foundational theory is semantic plurality. Variants differ in event queues, step semantics, priority, run-to-completion, concurrency, and history. Statecharts can be translated into transition systems, but the resulting semantics must be declared explicitly. They are therefore an effective authoring and presentation notation rather than the canonical mathematical substrate.

8. Event Structures

Event structures represent computations as event occurrences constrained by causality, consistency, enabling, and conflict. They distinguish true concurrency from mere nondeterministic interleaving and make partial-order traces primary. This is valuable for interaction because logs often impose a total order that is not causally justified, and because distributed or collaborative episodes contain independent events.

Event structures also support counterfactual questions about which events depend on which prior events and which alternatives exclude one another. These relations are closer to the interaction paper's recurrent-constraint account than a flat sequence is.

However, event structures are less direct for mutable data-rich state, continuous quantities, reward, partial observation, and online estimation. Probabilistic and game extensions exist, but tool ecosystems are narrower. Event structures should be available as a causal-concurrency semantics derived from or linked to the transition kernel.

9. MDPs and POMDPs

A Markov decision process represents states, actions, transition probabilities, rewards, and policies. It directly supports stochastic behavior, expected outcomes, controllability through action, and strategy evaluation. A partially observable MDP adds observations and belief states, making it especially relevant to the distinction between actual state, projection, observation, believed actions, and expected consequences.

These formalisms are the strongest candidates for opportunity, strategy, adaptation, and prediction. A projection can be represented partly through the observation kernel and available action function; belief mediates action selection; options can represent temporally extended policies.

Their standard assumptions create limits. The Markov property may require state augmentation. A single optimizing agent and scalar reward may misrepresent collaborative, institutional, or interpretive interaction. Authorization, signaling, semantics, and measurement validity are not intrinsic. POMDP observations are probabilistic signals, not the complete concept of representation or projection. MDP/POMDP semantics should therefore be an optional decision layer over the interaction kernel, not the universal ontology.

10. Category-Theoretic Process Models

Category theory provides objects, morphisms, composition, identity, products, monoidal structure, functors, and natural transformations. In process-oriented applications, objects can represent interfaces or systems and morphisms can represent processes. Symmetric monoidal categories support parallel composition and diagrammatic reasoning. Functors can relate different semantic models while preserving composition.

This makes category theory valuable for expressing the architecture of the research program: multiple views of one interaction model, compositional contracts, translations between Petri nets and execution categories, and preservation claims across projections. It is particularly suited to stating when a translation is structure preserving.

Category theory does not by itself choose the empirical entities, transition probabilities, observation model, or measurement semantics. Treating it as the primary model would move essential commitments into the choice of category. It should therefore serve as a meta-level compositional and translation framework, not the sole executable semantics.

11. Multi-Agent and Game-Based Models

Multi-agent systems represent multiple agents with local states, actions, observations, protocols, knowledge, goals, and sometimes utilities. Concurrent game structures add strategic choice and adversarial or cooperative outcomes. Epistemic and temporal logics can express what agents know and what coalitions can enforce.

These models directly address distributed agency and asymmetric opportunity. They can represent authorization, imperfect information, joint action, coordination, negotiation, and strategic interaction better than single-agent MDPs.

Their limitation is heterogeneity: multi-agent modeling is a family of frameworks rather than one semantics. Agent identity, rationality, knowledge, communication, and utility assumptions vary widely. The recommended foundation should permit multiple participant policies and local observation models without presupposing ideal rational agents or equilibrium behavior.

12. Comparative Result

No single candidate dominates across all obligations. LTSs lead on minimality, operational clarity, verification, and equivalence. Process algebras lead on native composition and observational substitutability. Event structures and Petri nets lead on true concurrency, causality, and resource flow.

Statecharts lead on practical hierarchical specification. POMDPs lead on partial observability, belief, policy, and stochastic decision. Category theory leads on compositional translation across models.

The comparison rejects two alternatives. A monolithic Cartesian-state POMDP can encode nearly every distinction but makes ontology, authority, representation, measurement, and composition opaque. A purely process-algebraic foundation captures interaction structure but leaves empirical uncertainty, belief, and data-model alignment as external additions. The minimal defensible solution is therefore a layered semantic profile with one canonical kernel and formally related views.

13. Recommended Formal Foundation

The recommended kernel is an observation-enriched probabilistic labeled transition system (OEPLTS):

$M = \langle N, S, A, T, I, F, C, P, X, O, Y, R \rangle$.

N is the set of participants and system components. S is a factored state space. A is a typed action/event vocabulary. $T(s, a, s')$ is a nondeterministic or probabilistic transition relation. I and F define episode initiation and completion. C defines effective capability. P defines authorization. X defines exposure through a projection. O maps hidden state and events to observer-accessible observations. Y maps executions to instrument records. R defines recovery or reversibility relations.

A participant-relative strategy is a history-, observation-, or belief-dependent policy π_i . Belief update is added when O is non-injective or stochastic. Rewards or criteria are optional and typed rather than mandatory. A coupling relation K identifies cross-component dependencies by which an event or state change of one component alters another component's enabled transitions, transition probabilities, observations, or value-relevant futures.

The kernel has three associated semantics. Its unfolding into an event structure provides causal and concurrent semantics. Its composition through process-algebraic operators provides modular semantics and behavioral equivalences. Its decision enrichment into an MDP, POMDP, stochastic game, or decentralized POMDP provides policy and uncertainty semantics. Petri-net or workflow projections represent resource and completion structure. Category-theoretic functors can state preservation across these views.

Empirical linkage is explicit: an execution ω generates an observation trace through O and an instrument record through Y . Segmentation and attribution functions turn records into behavioral events under a measurement contract. This prevents formal traces from being equated automatically with observed behavior.

14. Formal Semantics

An execution is a finite or infinite alternating sequence $s_0, a_0, s_1, a_1, \dots$ satisfying T . An interaction episode is an execution segment bounded by I and F in which the declared coupling criterion K is satisfied. Reciprocity is tested over the causal dependency graph, not merely by the presence of actions from multiple actors.

The executable set for participant i at state s is the intersection of domain-valid, capable, permitted, and enabled actions. Exposure and signaling are additional projection-relative predicates and need not coincide with executability. Believed availability is defined over the participant's belief state and may include false positives or omit executable actions.

Composition is defined by synchronized product over shared labels or explicit channels, with hiding and relabeling. Independent transitions may interleave operationally but are retained as concurrent in the event-structure unfolding. Refinement is typed: trace inclusion, simulation, bisimulation, probabilistic bisimulation, testing equivalence, opportunity preservation, or measurement-relative equivalence must be selected explicitly.

The model is implementation independent. A GUI, CLI, API, conversational agent, robot, or collaborative workflow is represented through different projection, observation, and action mappings over a declared domain and transition semantics.

15. Analyzability and Empirical Suitability

The recommended kernel preserves access to mature algorithms: reachability, invariant checking, temporal-logic model checking, simulation and bisimulation checking, probabilistic model checking, controller synthesis, policy evaluation, and partial-order reduction. Factored state and modular composition are necessary to reduce state explosion.

Empirical suitability requires a different set of checks. Model states and labels must be operationally linked to records; hidden variables must not be treated as observed; timing and missingness must be represented; and parameter estimates must carry uncertainty. Conformance between traces and models should distinguish deviation, observation error, abstraction error, and model misspecification.

The formal model is therefore not itself a measurement instrument. It is a theory-constrained representation whose empirical use depends on a measurement contract, estimation procedure, and validation evidence.

16. Falsification and Threats to Validity

The recommendation should be rejected or narrowed if a single established formalism can represent all required distinctions natively, preserve modularity, support the necessary analyses, and connect to empirical records without ad hoc layers. It should also be rejected if the proposed mappings cannot be made semantics-preserving or if independent analysts cannot reproduce the ontology-to-model mapping.

A second threat is over-formalization. Human interaction boundaries, intentions, semantics, and social rules may be contested or emergent. Formal precision does not eliminate uncertainty about what should be modeled. A third threat is state explosion and undecidability: increased expressiveness can destroy analyzability. A fourth is false realism: a model can be internally coherent while omitting institutional, environmental, or representational variables that generated observed behavior.

The foundation therefore makes scope conditions explicit. It is strongest for computationally mediated episodes with declared operations, participants, records, and boundaries. Open-ended cultural activity may require qualitative and historical explanation alongside formal modeling.

17. Conclusion

The minimum adequate foundation is not a new universal calculus. It is a conservative layered architecture centered on an observation-enriched probabilistic labeled transition system. The LTS kernel supplies state, transition, execution, reachability, and verification. Process algebra supplies composition and typed behavioral equivalence. Event structures or Petri nets supply concurrency, causality, and resource flow. POMDP or game enrichments supply partial observation, belief, policy, and stochastic strategy. Measurement mappings connect executions to records and claims. Category theory can organize translations and preservation across these views.

This recommendation preserves the strongest existing mathematics while matching the conceptual distinctions already established. It also leaves open an empirical test: if the layered model does not improve specification, comparison, explanation, verification, or prediction relative to simpler baselines, the additional structure should be removed.

18. Reusable Artifacts

A. Formal Ontology Mapping

Concept	Formal interpretation
Interaction episode	Bounded execution segment satisfying initiation, completion, and coupling predicates.
Participant	Typed component or agent with local state, actions, and observation access.
Domain state	Facts determining transition semantics independent of presentation.
Projection state	Representation and access-path configuration exposed to a participant.
Authority state	Roles, permissions, and policy conditions.
Observation	Participant-accessible symbol or signal generated from hidden state/event.
Instrument record	Timestamped or structured record emitted by the measurement channel.
Action/event	Typed occurrence capable of participating in a transition.
Transition	Possible or probabilistic state change labeled by an event.
Coupling	Cross-component dependency that changes possible or probable future trajectories.
Opportunity profile	Typed predicates over validity, capability, permission, enablement, exposure, belief, reachability, and recovery.
Strategy/policy	History-, observation-, or belief-dependent

	organization of action selection.
Execution	A path or partial-order configuration licensed by the model.
Behavioral trace	Attributed event structure derived from records under an observation and measurement contract.
Equivalence	Declared preservation relation over selected observables and transformations.

B. Model Comparison Matrix

Formalism	Native strengths	Uncertainty	Composition	Empirical fit	Equivalence	Recommended role
LTS	State, transitions, reachability, traces, temporal properties	Weak without extension	Strong	Strong	Moderate	Kernel
Petri net	Concurrency, synchronization, resources, invariants	Moderate via stochastic variants	Moderate	Strong for process logs	Moderate	Concurrent/resource view
Process algebra	Composition, communication, hiding, equivalence	Moderate via variants	Very strong	Moderate	Strong	Composition/equivalence layer
Statechart	Hierarchy, orthogonality, reactive lifecycle	Weak to moderate	Moderate	Strong for engineering models	Moderate	Authoring notation
Event structure	Causality, conflict, true concurrency	Moderate via extensions	Strong	Moderate	Strong	Causal-concurrency view
Workflow net	Completion, workflow soundness, process flow	Moderate	Moderate	Very strong	Moderate	Workflow specialization
MDP	Stochastic transitions, rewards, policies	Strong	Moderate	Strong with data	Moderate	Decision enrichment
POMDP	Observation, belief, policy under uncertainty	Very strong	Moderate	Strong but estimation-heavy	Moderate	Epistemic/strategy enrichment
Category theory	Composition and translations between models	Depends on chosen category	Very strong	Weak alone	Abstractly strong	Meta-level architecture
Multi-agent/game	Local observation, joint action, strategic ability	Strong in stochastic variants	Moderate	Moderate	Strong for strategic equivalence	Participant enrichment

C. Expressiveness Mapping

Required construct	Recommended representation
Episode boundaries	Predicates I and F over execution states/events
Reciprocal coupling	Dependency relation K over components, transitions, observations, or enabled sets
Concurrency	Event-structure unfolding or Petri-net projection; partial-order reduction
Opportunity	Typed state/action predicates C, P, E, X plus reachability and recovery
Projection	State-dependent mapping from domain model to exposed actions and observations
Measurement	Channel Y from execution to records plus segmentation/attribution functions
Uncertainty	Probabilistic transition and observation kernels;

	interval or imprecise variants where needed
Strategy	Policies over histories, observations, or beliefs; options for temporal extension
Composition	Synchronized product, parallel operator, hiding, relabeling, interface contracts
Similarity/equivalence	Trace, simulation, bisimulation, testing, probabilistic, opportunity, and measurement- relative relations

D. Recommended Formal Profile

Canonical kernel: observation-enriched probabilistic labeled transition system.

Composition semantics: process-algebraic synchronized product, hiding, relabeling, recursion, and typed equivalence.

Concurrency semantics: event-structure unfolding or Petri-net projection when causal independence and resources matter.

Decision semantics: MDP, POMDP, stochastic-game, or decentralized enrichment only when uncertainty, policy, or strategic choice is part of the explanatory target.

Empirical semantics: observation channel and instrument-record channel governed by the Measurement Contract and Behavioral Evidence Contract.

Translation semantics: explicit structure-preserving mappings between views, optionally formalized as functors.

19. References

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